Project - High Level Design

On

Dockerized Construction Services

Course Name: DevOps

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Introduction

This document presents the system design for the Dockerized Construction Service, developed as part of a DevOps-based project. The project demonstrates a complete containerization workflow using Docker along with CI/CD automation.

The application represents a Construction Planning Service integrated with GenAI capabilities, where intelligent modules assist in planning, scheduling, and resource estimation using AI-driven insights.

The DevOps pipeline ensures the application is:

- Consistently packaged
- Easily deployable
- Scalable across environments
- Secure and optimized

1.1. Scope of the document

This document includes:

- Docker-based containerization strategy
- Multi-stage Docker build optimization
- Docker Compose orchestration
- CI/CD pipeline using GitHub Actions / Jenkins
- Environment and dependency management
- Security and performance considerations
- API catalogue and service endpoints

1.2. Intended Audience

This document is intended for:

- DevOps Engineers
- Backend Developers
- AI Engineers
- Evaluators
- Future Maintainers

1.3. System overview

The system consists of two main repositories:

Repository 1 – Construction Service(Application)

Contains:

- Backend logic (construction planning)
- GenAI Modules (task suggestions, cost estimation)
- APIs and business logic

Repository 2 – DevOps Infrastructure

Contains:

- Docker file
- Docker compose
- CI/CD pipeline
- Environment configs

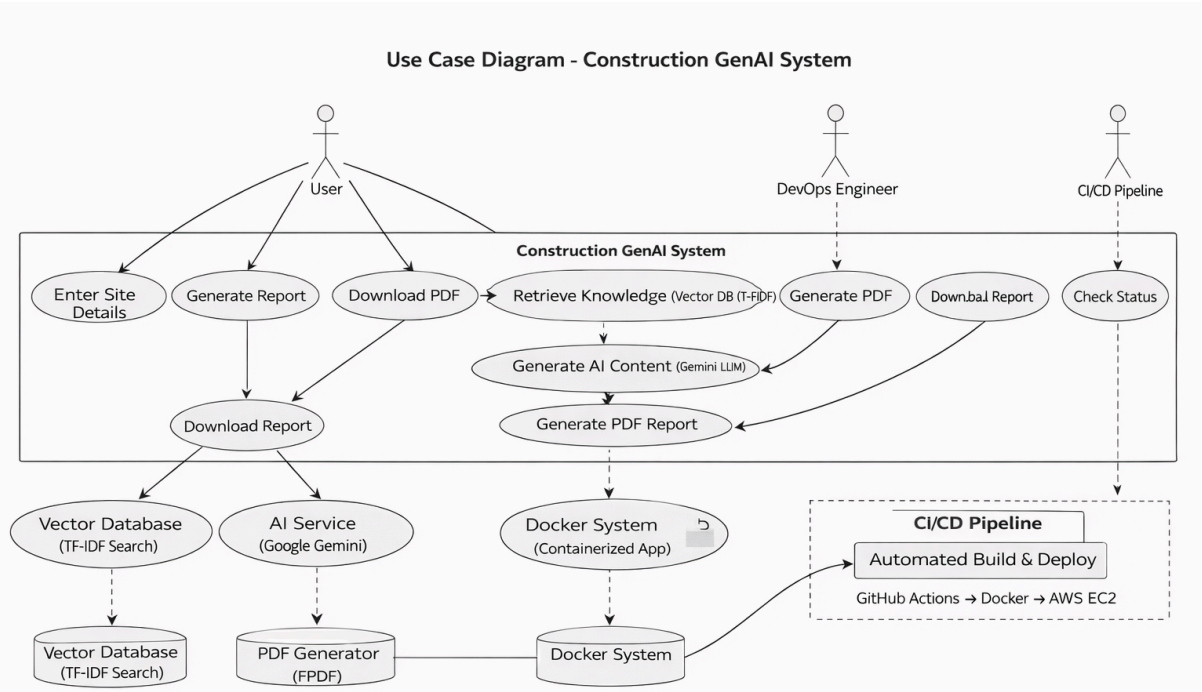
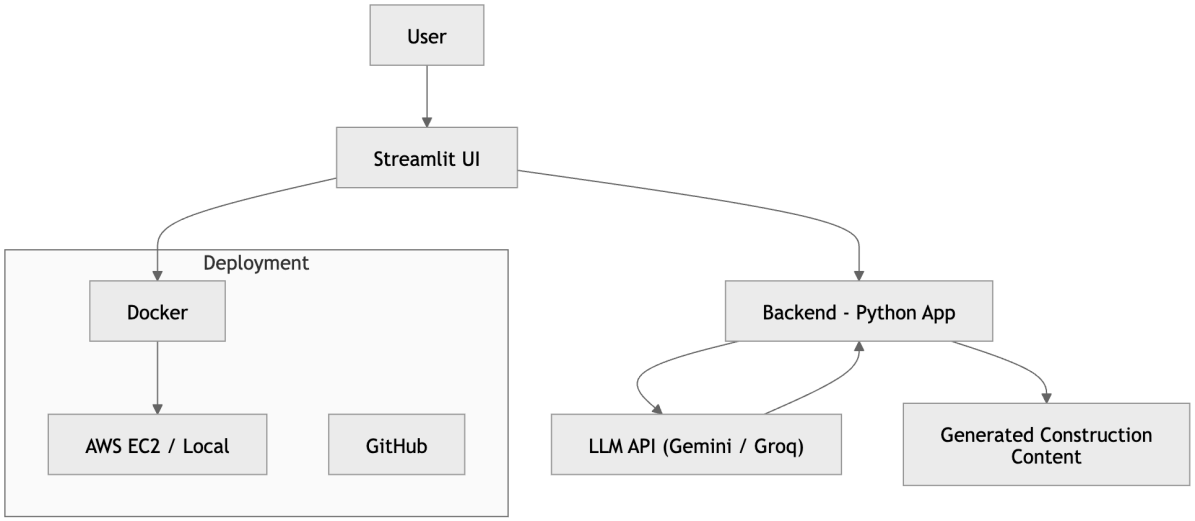


Figure 1: Use Case Diagram

High-Level Workflow

- Developer pushes code → GitHub
- CI/CD pipeline triggers
- Docker image is built using multi-stage build
- Image is stored in Docker Hub
- Docker Compose deploys services
- Application becomes accessible

2. System Design



2.1. Application Design

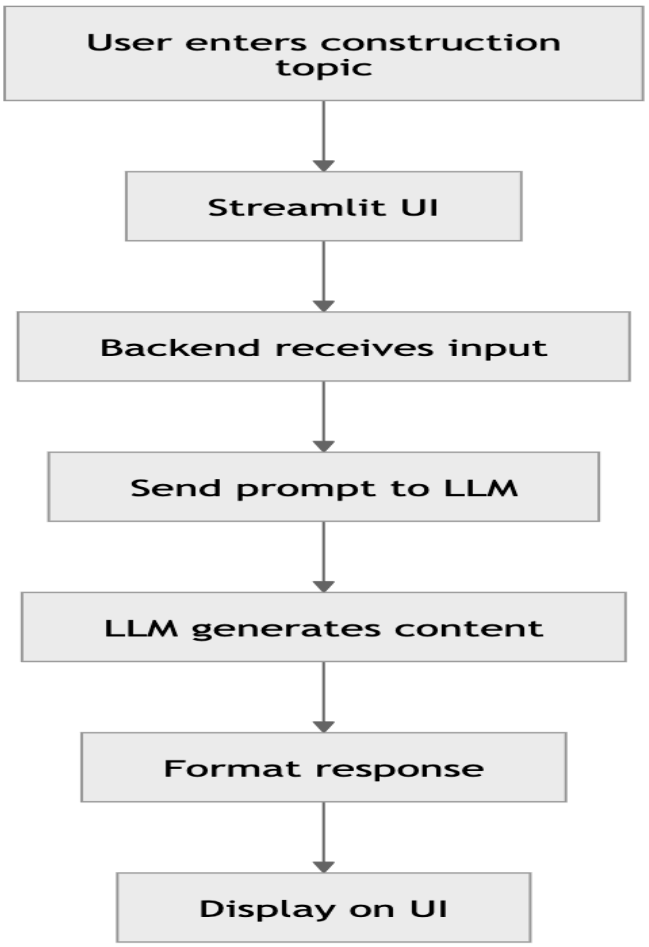
The **Construction GenAI Application** is designed as a modular system that integrates user input, AI processing, and report generation.

- The **frontend** is built using Streamlit, allowing users to enter construction site details and request reports.
- The **backend logic** processes user input, applies prompt engineering, and interacts with the AI model.
- A **vector database (TF-IDF)** is used to retrieve relevant documents from the knowledge base to enhance accuracy (RAG approach).
- The system integrates with **Google Gemini** to generate structured and context-aware construction reports.
- The generated content is converted into a **PDF report using FPDF**.
- The entire application is **containerized using Docker** and deployed on AWS EC2 through a **CI/CD pipeline (GitHub Actions)**

2.2. Process Flow

The system follows a structured execution flow:

- User sends request (e.g., "Create construction plan")
- Backend processes input
- GenAI module generates intelligent suggestions
- Backend validates and structures output
- Response returned to user



2.3. Information Flow

The flow of information in the system is as follows:

User → Frontend → Backend → GenAI Module → Backend → Frontend → User

2.4. Components Design

- **Component 1 – Docker file**

Property	Value
Base Image	python:3.11-slim
Build Type	Multi-stage build
Working Directory	/app
Exposed Port	8000
Entry Point	python app.py

- **Component 2 – Docker Compose**

Service	Description
app	Main construction service
db	Database(PostgreSQL/MySQL)
cache	Redis (optional)

- **Component 3 – CI/CD Pipeline**

(using GitHub Actions / Jenkins)

Step	Action
1	Checkout code
2	Install dependencies
3	Build Docker image
4	Run tests
5	Push image to Docker Hub
6	Deploy using Docker Compose

- **Component 4 – Docker hub Registry**

Action	Purpose
Push Image	Store container image
Pull Image	Used during deployment

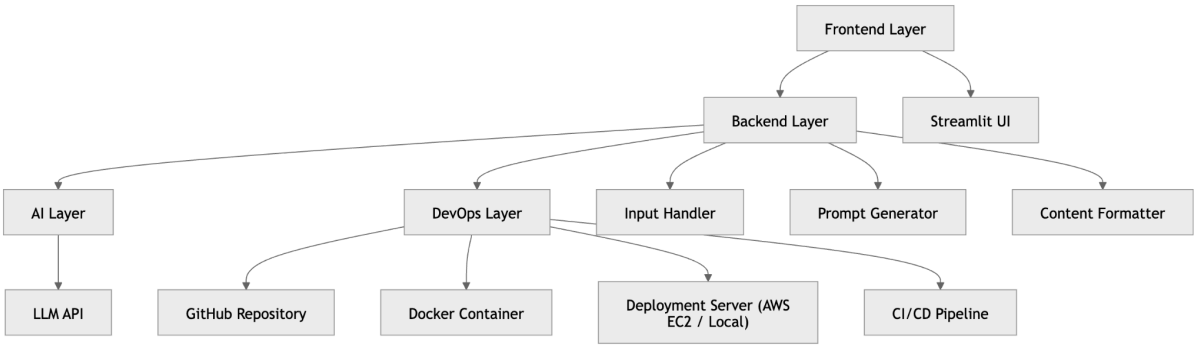


Fig 2.4 Component Diagram

2.5. Key Design Considerations

1. Multi-stage Docker build

- Reduces image size
- Improves security
- Removes unnecessary files

2. Separation of Concerns

- Application code and DevOps configs separated
- Enables parallel development

3.Environment Consistency

- Same container runs in dev, test, prod
- Eliminates "works on my machine" problem

4.GenAI Integration

- AI assists in construction planning
- Keeps logic modular and scalable

5.Security

- Minimal base images
- No secrets stored in image
- Environment variables used

6. Cost Efficiency

- Lightweight containers
- Minimal infrastructure usage

2.6. API Catalogue

1. Internal APIs

Endpoint	Method	Purpose
/generate-report	POST	Generate construction report using AI
/process-prompt	POST	Prepare input prompt for LLM
/llm-response	POST	Get response from Gemini LLM
/generate-pdf	POST	Convert report into PDF
/download-report	GET	Download generated report
/load-data	GET	Load knowledge base documents

External APIs

1. Google Gemini API

API	Method	Purpose
Gemini Generate Content API	POST	Generates construction reports based on user input and prompts

Authentication:

- API key via environment variables

3. Interfaces

The system provides both User Interface and External Interfaces for smooth interaction.

1. User Interface

- Built using Streamlit
- Allows users to:
    - Enter construction site details
    - Generate AI-based reports
    - View and download PDF reports

2. External Interfaces

- Google Gemini API
  Used for generating AI-based construction content
- File System (Data Folder)
  Stores knowledge base documents for retrieval

3. System Interfaces

- Vector Database (TF-IDF)
  Retrieves relevant documents for context-aware generation
- PDF Generator (FPDF)
  Converts generated content into downloadable reports

4. Data Design

The Data Design of the Construction GenAI system focuses on handling user inputs and AI-generated outputs efficiently without relying on persistent storage. The system follows a stateless, API-driven architecture, where data is processed in real-time and returned to the user. The design ensures simplicity, fast processing, and secure handling of API interactions.

4.1 Data Model

The system uses a **dynamic data model**, where data is generated and processed during runtime rather than stored permanently.

Key Data Entities:

- **User Input**
    - Construction query (e.g., planning, estimation, safety)
    - Format: String

- **Processed Prompt**
  - Refined input sent to LLM
  - Includes structured instructions

4.2 Data Access Mechanism

The system accesses and transfers data through API-based communication and DevOps pipeline flow.

Data Flow:

- User → Streamlit UI → Backend → LLM API → Response → UI

Access Methods:

- **HTTP Requests**
  - Used for communication with LLM APIs
- **GitHub**
  - Stores application code
- **Docker**
  - Packages application
- **Cloud (AWS EC2 / similar)**
  - Hosts application

4.3 Data Retention Policies

The system follows a no persistent storage policy for simplicity and security.

Policies:

- No database storage implemented
- Data exists only during user session
- No user history tracking
- No sensitive data stored

4.4 Data Migration

Currently, **data migration is not required** due to the absence of a database.

Current State:

- Stateless system
- No stored data
- No versioned data migration

5. State and Session Management

- Managed using Streamlit session state
- Each user session:
  - Is isolated
  - Stores temporary inputs and outputs
- No persistent storage is used
- Execution results are:
  - Generated dynamically
  - Lost after session ends or refresh

6. Caching

- No caching mechanism implemented in current system
- Each request:
  - Triggers fresh LLM execution

7. Functional Requirements

The system must:

- Accept construction site details from user
- Process input and generate AI prompt
- Retrieve relevant data from knowledge base
- Generate report using AI (Gemini)
- Convert report into PDF
- Display report to user
- Allow download of PDF report

7.1 User Authentication & Authorization

- No authentication system implemented
- Users access application directly
- API keys (Gemini/Groq):
  - Entered at runtime via UI
  - Not stored in code or database

7.2 Infrastructure Configuration Management

- Configuration managed via:
  - Environment variables
  - .env files (if used locally)
- Key configurations:
  - API keys
  - Model parameters

- System follows:
  - **Code-based configuration**
  - Easy portability across environments

7.3 Provisioning Control & Monitoring

Provisioning:

- Application runs locally via:
  - streamlit run app.py

Monitoring:

- Basic monitoring via:
  - Console logs
  - Streamlit runtime logs

7.4 Dockerfile (Application Containerization)

Configuration:

Property	Value
Base Image	python:3.10/3.11
Working Directory	/app
Port	8501
Entry Point	streamlit run app.py

Key Steps:

- Install dependencies
- Copy project files
- Expose port 8501
- Run Streamlit server

7.5 GitHub Actions (Workflow Configuration)

Workflow Steps:

1. Checkout repository
2. Install dependencies
3. Run tests (if any)
4. Build Docker image
5. Push image to registry
6. Deploy (optional)

Secrets Used:

- API keys
- Docker credentials

7.6 CI/CD Strategy (Continuous Integration & Deployment)

Continuous Integration:

- Trigger: git push
- Actions:
  - Code validation
  - Build process
  - Dependency installation

Continuous Deployment (Optional Extension):

- Deploy to:
  - Docker container
  - Cloud (AWS)

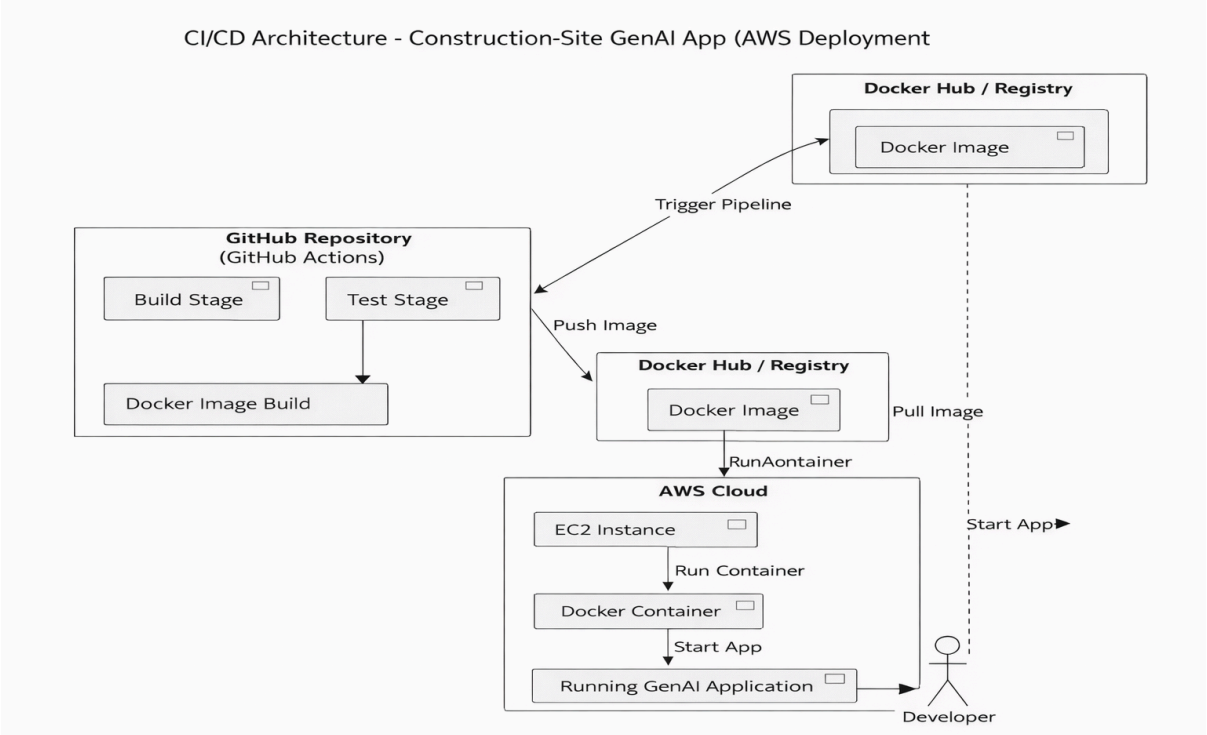


Figure 1: CI/CD Architecture Diagram

8. Non-Functional Requirements

- **Performance**: The system should generate construction plans within a reasonable response time, primarily dependent on LLM API latency.
- **Usability:** The user interface should be simple and intuitive, allowing users to input goals and view results without technical complexity.
- **Reliability:**The system should process each request independently to ensure consistent and predictable outputs without affecting other sessions.

9. References

- Docker Documentation
- GitHub Actions / Jenkins Docs
- GenAI API Documentation

10. GitHub

Github link - https://github.com/anshu2426/construction-genai